Module 14: Macroevolution - Key Points

Learning Objectives

By the end of this module, you should be able to:

1. State the biological species concept and its limits.
2. Explain reproductive isolation as reduced gene flow.
3. Classify examples as prezygotic or postzygotic barriers.
4. Compare allopatric and sympatric speciation.
5. Interpret speciation as population-level divergence over time.

Topics

- Species concepts
- Reproductive isolation
- Prezygotic barriers
- Postzygotic barriers
- Speciation patterns and phylogeny

Key Terms to Know

- **Species** - Lineage or group identified by a species concept.
- **Biological species concept** - Species defined by interbreeding and reproductive isolation.
- **Reproductive isolation** - Barriers that reduce gene flow between groups.
- **Prezygotic barrier** - Barrier before fertilization.
- **Postzygotic barrier** - Barrier after fertilization.
- **Allopatric speciation** - Speciation with geographic separation.
- **Sympatric speciation** - Speciation without geographic separation.
- **Phylogeny** - Evolutionary relationship history among lineages.

Core Contents

1. Species concepts define lineages using reproductive, morphological, or evolutionary criteria.
2. Reproductive isolation reduces gene flow between populations.
3. Prezygotic barriers prevent mating or fertilization before a zygote forms.
4. Postzygotic barriers reduce hybrid survival or fertility after a zygote forms.
5. Speciation can occur allopatrically or sympatrically and is represented in phylogenies.

Study Tips

1. Start with the module's central claim before memorizing vocabulary.
2. Use the same-numbered lab as the evidence check for the module.
3. Explain one mechanism in words, then redraw it as a simple diagram.
4. Answer retrieval questions without notes, then revise with the terms list.

Connected Lab

- `lab-14_macroevolution.md`

Generated Visuals

- Module 14: Macroevolution Evidence Map
- Module 14: Macroevolution Reasoning Sequence
- Module 14: Macroevolution Retrieval and Lab Check